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SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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SDC HACKATHON 2026

Technical & Tools Declaration

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1. Team & Project Overview

Team Name	UM IS ONE
Project / Solution Name	AI Skill Compass
Name (Leader)	ANG JI JIN
Email & Phone (Leader)	angjijin20071008@gmail.com & 017-823 9621
University Affiliation(s)	University Malaya
Live Prototype URL	https://sdc-domain-1-prototype.replit.app/
Code Repository URL (public / access granted to judges)	https://github.com/Codinggeniusnks/sdc-domain1-prototype

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

AI Skill Compass is a stateless Python 3.11 web application deployed on Replit Autoscale. The browser receives server-side-rendered HTML/CSS with lightweight vanilla JavaScript, while src/main.py exposes JSON REST endpoints for health, trend signals and profile evaluation.

User profile or CV text is normalised and passed to M1, the frozen hybrid skills-extraction model. Extracted skills are combined with M2 labour-market trend intelligence, then consumed by the role-matching, reskilling and SDG layers. M6 applies transparent alternative AI-adoption scenarios so results are shown as bounded estimates rather than a single deterministic future.

Live research-signal mode queries OpenAlex and arXiv with short timeouts; failures automatically fall back to a bundled dated snapshot. Model files, source crosswalks and scenario outputs are stored as versioned read-only local artefacts.

The deployed prototype does not require a database, user authentication, persistent user storage or a runtime generative-AI API. Replit hosts the web server and maps the application's environment PORT to the public service.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
Python 3.11 (incl. standard library: json, pathlib, hashlib, urllib, http.server)	Backend, AI/ML inference, data processing, scoring engines, file handling, hashing, API access; http.server.ThreadingHTTPServer provides the custom SSR web server and JSON REST API	PSF License
HTML5 / CSS3 / Vanilla JavaScript	Server-side-rendered interface, responsive UI and design system, progressive enhancement and API interactions	Open web standards (no third-party library)
scikit-learn	TF-IDF, Logistic Regression, multi-label classification and evaluation	BSD-3-Clause
sentence-transformers	Sentence embedding generation (MiniLM)	Apache 2.0
Hugging Face Transformers	Runtime dependency for the MiniLM embedding model	Apache 2.0
PyTorch	Runtime dependency for the MiniLM embedding model	BSD-3-Clause
NumPy	Numerical operations and model calculations	BSD-3-Clause
pandas	Structured dataset processing and analysis	BSD-3-Clause
joblib	Loading/saving trained scikit-learn artefacts	BSD-3-Clause
pip / requirements.txt	Dependency installation and version management	MIT

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
OpenAlex REST API (api.openalex.org) – OurResearch	Retrieves current scholarly/research signals on AI and workforce trends. Live mode with local snapshot fallback. Sends research query terms only; no CV/profile content.	Free
arXiv API (export.arxiv.org) – Cornell University	Retrieves recent research-publication signals. Live mode with local snapshot fallback. Sends research query terms only; no CV/profile content.	Free
OpenDOSM / data.gov.my Open API – Department of Statistics Malaysia	Labour-force and wage dataset acquisition/refresh during offline/periodic data preparation. No user data sent.	Free
RunPod API – RunPod	Optional support for separate batch-compute scripts. Not required by the deployed web server and not used by the current frozen production build. No user data sent.	Sponsor-provided
Note	No Gemini, OpenAI, Claude or other generative-AI API is called by the deployed application at runtime. No payment, mapping or messaging APIs are used.	–

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit Autoscale – Replit	Web application hosting and compute for the production/prototype deployment	Autoscale deployment (sponsor-provided); region managed by Replit
Replit Environment Variables / Secrets – Replit	Deployment configuration and optional credentials, kept outside source code	Included with Replit deployment (sponsor-provided)
RunPod – RunPod	Optional remote/batch compute capability supported by separate tooling; not required or used by the current deployed web application	Sponsor-provided credits
Local bundled storage	Versioned JSON, model and snapshot artefacts bundled read-only with the deployment (no database)	N/A (bundled with deployment)
GitHub – GitHub, Inc.	Source code repository and version control	Free

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
M1 Hybrid Skills Extraction Model – H_blend_0.70A_0.30C1_W1.0	Team UM IS ONE	Self-trained hybrid multi-label classifier; extracts skills from profile/CV text (≈70% lexical + 30% MiniLM semantic probability)	Team-owned
Candidate A component – A_base_C2.5_W1.0	Team UM IS ONE (built with scikit-learn)	Self-trained: word + character TF-IDF with Logistic Regression for lexical skill recognition	Team-owned; scikit-learn BSD-3-Clause
all-MiniLM-L6-v2	Hugging Face / Sentence Transformers	Pre-trained 384-dimensional sentence embedding encoder for semantic skill recognition	Apache 2.0
Candidate C1 component – MiniLM embeddings + Logistic Regression	Team UM IS ONE	Self-trained classification head for semantic multi-label skill classification	Team-owned
M2 Trend Intelligence Engine – v1.1.1	Team UM IS ONE	Self-created deterministic evidence-fusion/scoring engine combining labour-market, research, policy, wage and AI-exposure signals	Team-owned
M6 Future Workforce Scenario Engine – v1.1.0	Team UM IS ONE	Self-created deterministic scenario model (Reference, Moderate Copilot, Fast Agentic scenarios)	Team-owned

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Dataset v5 – Clean Multisource Skills Dataset	Team-created from the sources declared below	Self-collected (derived; small synthetic portion)	Team-owned; inherits source terms	N
masco_mycol_occupations.json	MASCO 2020 / TalentCorp MyCOL – emasco.mohr.gov.my, talentcorp.com.my/mycol	Public	Public government publication; attributed	N
masco_mycol_expanded.json	Team-curated derivative of MASCO/MyCOL information (not a verbatim official dataset)	Self-collected (derived)	Team-owned; source attributed	N
MASCO 2020	Ministry of Human Resources / JTKSM – emasco.mohr.gov.my, jtksm.gov.my	Public	Public government publication; attributed	N
MyMahir Malaysia Critical Occupations List (MyCOL)	TalentCorp – talentcorp.com.my/mycol	Public	Public publication; attributed	N
TalentCorp workforce impact studies	TalentCorp – talentcorp.com.my	Public	Public publication; cited	N
OpenDOSM / DOSM Labour & Wage Data	open.dosm.gov.my, api.data.gov.my	Public	CC BY 4.0	N

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
ILO Working Paper 140 (2025) and Working Paper 96 (2023)	International Labour Organization – ilo.org	Public	ILO publication; cited	N
openalex_works.json / OpenAlex research corpus	openalex.org, api.openalex.org	Public	CC0	N
esco_occupations.json / ESCO	European Commission – esco.ec.europa.eu	Public	ESCO free reuse with attribution	N
frameworks_expanded.json	Team-curated competency corpus based on declared public sources and internally authored skill objectives (not a verbatim official dataset)	Self-collected (derived)	Team-owned; sources attributed	N
Malaysian Policy Evidence Corpus	MOSTI AIGE, Bursa Malaysia sustainability frameworks, PDPA Amendment Act 2024, MyDIGITAL/4IR, MITI National Semiconductor Strategy	Public	Public government/regulator publications; cited	N
data/signals_snapshot.json	Team-generated dated fallback snapshot from public evidence sources	Self-collected (derived)	Team-owned; inherits source terms	N
jobstreet_postings.json	https://www.jobstreet.com/	Self-collected	Team-owned	N

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☒ Yes ☐ No

If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
OpenAI ChatGPT	Architecture discussion, technical review, research synthesis, debugging, documentation and presentation support	Moderate
OpenAI Codex	Code implementation, code review, debugging and repository-development assistance	Major
Google Antigravity	Agent-assisted implementation, testing, debugging and development workflow	Major
Gemini 3.8 Flash (via Antigravity)	Coding, local implementation tasks and technical analysis	Moderate
Gemini 4	Development assistance and higher-level technical reasoning where used	Minor
Claude Opus 5.5 (via Antigravity)	Architecture/methodology review, debugging and implementation assistance	Moderate

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

These tools were engineering assistants only and are not runtime dependencies of the deployed application. Every AI-generated suggestion was read and understood by a team member before being accepted, modified or rejected. Code changes were run and tested locally, model outputs were checked against held-out evaluation data, and data sources and labour-market claims were verified against the original publications. The team takes full responsibility for all code, data and text in the submission.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☒ Yes ☐ No

If yes, list them below:

Sponsor	**Resource (API / credits / platform / data)**	**How It Was Used in the Project**
Replit	Platform / cloud credits (Replit Autoscale, Secrets)	Hosts and serves the deployed Python web application (prototype/production deployment) and stores deployment configuration outside source code
RunPod	Cloud GPU compute / API credits	Remote batch compute for separate development scripts; not required by, and not used in, the current frozen production build

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Replit is essential to how the prototype is currently hosted and accessed by judges, but it is replaceable: the application is a stateless Python 3.11 server using only the standard-library HTTP server and bundled read-only artefacts, so it can run on any Python host or container platform without code changes. RunPod is not essential; it supported optional batch-compute tooling only, and the deployed web application runs entirely without it.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

The prototype may process user-supplied CV/profile text, skills, experience and career interests. It does not require payment data, government identifiers or account credentials. In the current deployment there is no persistent user database or authentication layer; profile input is processed for the request/session and is not intentionally stored as a long-term user record.

Raw user profile content is not sent to Gemini, ChatGPT, Claude or another generative-AI API at runtime. Live OpenAlex/arXiv requests contain only research/trend queries, not the user's CV. API credentials, where applicable, are stored in environment variables rather than source code. Logs should exclude raw CV text and secrets.

Bundled labour-market datasets contain public occupational, wage, policy and research information, not personal records.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

The main limitations are model and evidence coverage rather than generative hallucination. M1 is currently English-focused and may miss uncommon phrasing, emerging skills or Bahasa Malaysia/Chinese CV content. Its confidence scores are used with human-confirmation thresholds rather than treated as calibrated probabilities.

Some training data is team-curated and a small portion is synthetic, so generalisation is not guaranteed. OpenAlex research momentum can be noisy for low-base topics. Malaysian wage evidence is available at broad MASCO major-group level and must not be presented as role-specific salary estimates.

Live OpenAlex/arXiv services may be unavailable or rate-limited; the application falls back to dated snapshots. M6 scenarios are transparent what-if estimates, not forecasts of actual employment outcomes, and results depend on the quality and freshness of upstream labour-market evidence.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- ☒ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☒ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☒ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☒ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☒ Any use of official SDC sponsor resources has been declared in Section 5.
- ☒ The prototype described here is accessible to judges through the links provided in Section 1.
- ☒ All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.



October 24 2026